# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## CLASS 9 — MATHS · MEDIUM LEVEL MCQ TEST SET 3

**Full Syllabus (Chapters 1–14)** | RBSE Pattern | Session 2026–27

**Time: 1 Hour** Name: _____ Roll No.: _____ **Maximum Marks: 50**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. Medium level: multi-step questions.

## MULTIPLE CHOICE QUESTIONS — 50 × 1 = 50

*Choose the correct option and write it in your answer book.*

**1.** $-9 + (-4) \times (-5) =$ 1

**(a)** 13 **(b)** 12 **(c)** 11 **(d)** 10

**2.** If $x + y = 12$ and $x - y = 4$, then $xy =$ 1

**(a)** 33 **(b)** 31 **(c)** 32 **(d)** 34

**3.** $AB$ is a diameter and $C$ lies on the circle with $\angle ABC = 24^\circ$. Then $\angle BAC =$ 1

**(a)** $76^\circ$ **(b)** $56^\circ$ **(c)** $66^\circ$ **(d)** $86^\circ$

**4.** A wheel of diameter $70$ cm (take $\pi = \frac{22}{7}$) makes $50$ revolutions. The distance covered (in metres) is: 1

**(a)** 112 **(b)** 109 **(c)** 110 **(d)** 111

**5.** Three coins are tossed. The probability of getting at most one head is: 1

**(a)** $\frac{1}{2}$ **(b)** $\frac{1}{3}$ **(c)** $\frac{1}{4}$ **(d)** $\frac{2}{3}$

**6.** The value of the 20th term minus the 10th term of the AP $3, 10, 17, 24, \ldots$ is: 1

**(a)** 70 **(b)** 77 **(c)** 80 **(d)** 63

**7.** The number of integers from 1 to 50 that have an odd number of factors is: 1

**(a)** 9 **(b)** 8 **(c)** 6 **(d)** 7

**8.** A shop earns a profit of $9\%$ on sales of ₹3000 and $24\%$ on sales of ₹2000. The overall profit percentage on total sales is: 1

**(a)** $14\%$ **(b)** $17\%$ **(c)** $16\%$ **(d)** $15\%$

**9.** By Euclid's algorithm, gcd(453, 384) is: 1

**(a)** 5 **(b)** 3 **(c)** 4 **(d)** 2

**10.** In $\triangle ABC$, D and E are the midpoints of AB and AC and $DE = 30$ cm. Then BC is: 1

**(a)** 62 cm **(b)** 90 cm **(c)** 64 cm **(d)** 60 cm

**11.** A line with slope $6$ passes through $(0, 7)$. Its $y$-intercept is: 1

**(a)** 9 **(b)** 8 **(c)** 6 **(d)** 7

**12.** A pyramid has a square base of side $12$ cm and height $24$ cm. Its volume is: 1

**(a)** $1152$ cm$^3$ **(b)** $1384$ cm$^3$ **(c)** $1036$ cm$^3$ **(d)** $1268$ cm$^3$

**13.** The square of the distance between $(-5, 3)$ and $(2, -1)$ is: 1

**(a)** 67 **(b)** 62 **(c)** 70 **(d)** 65

**14.** If $x = 5$ is a zero of $6x + m$, then $m =$ 1

**(a)** -28 **(b)** -30 **(c)** -29 **(d)** -31

**15.** Which of the following has a terminating decimal expansion? 1

**(a)** $\frac{4}{33}$ **(b)** $\frac{11}{39}$ **(c)** $\frac{8}{15}$ **(d)** $\frac{21}{70}$

**16.** $(6x - 5y)^2 =$ 1

**(a)** $36x^2 + 60xy + 25y^2$ **(b)** $36x^2 - 60xy + 25y^2$ **(c)** $36x^2 + 25y^2$ **(d)** $36x^2 - 30xy + 25y^2$

**17.** The length of a chord of a circle of radius $10$ cm lying $8$ cm from the centre is (in cm): 1

**(a)** 13 **(b)** 12 **(c)** 14 **(d)** 11

**18.** The length of the arc of a sector of angle $240^\circ$ in a circle of radius $21$ cm (take $\pi = \frac{22}{7}$) is (cm): 1

**(a)** 88 **(b)** 89 **(c)** 90 **(d)** 87

**19.** Two dice are rolled together. The probability that the sum is 9 is: 1

**(a)** $\frac{1}{9}$ **(b)** $\frac{2}{9}$ **(c)** $\frac{8}{9}$ **(d)** $\frac{1}{10}$

**20.** In an AP, $t_1 = 50$ and $t_9 = 26$. The common difference is: 1

**(a)** $-3$ **(b)** $-1$ **(c)** $-4$ **(d)** $-2$

**21.** The converse of "If a quadrilateral is a square, then it is a rectangle" is false because: 1

**(a)** $x = -3$ gives $x^2 = 9$ but $x \neq 3$

**(b)** two angles of $60^\circ$ in an equilateral triangle are equal but not vertically opposite

**(c)** a kite that is not a rhombus has perpendicular diagonals

**(d)** a $4$ cm by $6$ cm rectangle is not a square

**22.** The average of 37 numbers is $86$. When a number $48$ is included, the average of all 38 numbers becomes: 1

**(a)** $86$ **(b)** $85$ **(c)** $87$ **(d)** $84$

**23.** Using Euclid's subtraction algorithm for gcd$(294, 245)$, the number of subtraction steps needed is: 1

**(a)** $6$ **(b)** $7$ **(c)** $8$ **(d)** $5$

**24.** The angles of a quadrilateral are in the ratio $5 : 2 : 6 : 2$. The largest angle is: 1

**(a)** $144^\circ$ **(b)** $164^\circ$ **(c)** $154^\circ$ **(d)** $134^\circ$

**25.** The slope of the line $3x + 6y = 15$ is: 1

**(a)** $\frac{3}{2}$ **(b)** $-\frac{3}{2}$ **(c)** $-\frac{1}{2}$ **(d)** $\frac{1}{2}$

**26.** The radius of a sphere is multiplied by $4$. Its surface area becomes ___ times: 1

**(a)** 16 **(b)** 64 **(c)** 4 **(d)** 8

**27.** The midpoint of $(-5, 4)$ and $(2, 9)$ is: 1

**(a)** $(-3.5,\ -2.5)$ **(b)** $(-1.5,\ 7.5)$ **(c)** $(-1,\ 6.5)$ **(d)** $(-1.5,\ 6.5)$

**28.** The slope of the line through $(0, -2)$ and $(4, 10)$ is: 1

**(a)** 3 **(b)** 2 **(c)** 4 **(d)** 5

**29.** The number of decimal places in the decimal expansion of $\frac{13}{250}$ is: 1

**(a)** 3 **(b)** 5 **(c)** 2 **(d)** 4

**30.** $(3x+2)(2x+7) =$ 1

**(a)** $6x^2+17x+14$ **(b)** $6x^2+26x+14$ **(c)** $6x^2+25x+14$ **(d)** $6x^2+25x-14$

**31.** A chord of length $16$ cm lies $6$ cm from the centre of a circle. The radius (in cm) is: 1

**(a)** 12 **(b)** 10 **(c)** 9 **(d)** 11

**32.** A triangle has area $84$ cm$^2$ and one side $21$ cm. The altitude on this side (cm) is: 1

**(a)** 8 **(b)** 9 **(c)** 10 **(d)** 7

**33.** Two dice are rolled together. The probability of getting a sum greater than 9 is: 1

**(a)** $\frac{1}{7}$ **(b)** $\frac{5}{6}$ **(c)** $\frac{1}{3}$ **(d)** $\frac{1}{6}$

**34.** In an AP, $d=-2$ and $t_8=16$. The first term is: 1

**(a)** $31$ **(b)** $29$ **(c)** $32$ **(d)** $30$

**35.** Which of the following is a counterexample to the statement "If a number is divisible by 4 and by 6, then it is divisible by 24"? 1

**(a)** $72$ **(b)** $48$ **(c)** $12$ **(d)** $24$

**36.** The average of 31 numbers is $66$. If the number $156$ is removed, the average of the remaining 30 numbers is: 1

**(a)** $63$ **(b)** $65$ **(c)** $64$ **(d)** $62$

**37.** Using Euclid's subtraction algorithm for gcd$(448, 176)$, the number of subtraction steps needed is: 1

**(a)** $11$ **(b)** $8$ **(c)** $9$ **(d)** $10$

**38.** One angle of a parallelogram exceeds another adjacent angle by $80^\circ$. The smaller angle is: 1

**(a)** $45^\circ$ **(b)** $50^\circ$ **(c)** $90^\circ$ **(d)** $60^\circ$

**39.** The points $(-2,-1)$, $(-1,-4)$ and $(0,k)$ are collinear. The value of $k$ is: 1

**(a)** $-7$ **(b)** $-8$ **(c)** $-6$ **(d)** $-10$

**40.** The edge of a cube is multiplied by $6$. Its volume becomes ___ times: 1

**(a)** 18 **(b)** 216 **(c)** 6 **(d)** 36

**41.** One end of a diameter of a circle is $(-3,6)$ and the centre is $(0,3)$. The other end is: 1

**(a)** $(0,3)$ **(b)** $(-3,0)$ **(c)** $(3,0)$ **(d)** $(4,0)$

**42.** The $x$-intercept of the line $y=6x-12$ is: 1

**(a)** 3 **(b)** 4 **(c)** 2 **(d)** 1

**43.** $0.\overline{54}$ in lowest terms is: 1

**(a)** $\frac{7}{11}$ **(b)** $\frac{6}{11}$ **(c)** $\frac{5}{11}$ **(d)** $\frac{1}{2}$

**44.** If $a-b=5$ and $ab=24$, then $a^2+b^2=$ 1

**(a)** 73 **(b)** 75 **(c)** 72 **(d)** 74

**45.** In a circle with centre $O$, chord $AB$ has $\angle AOB=84^\circ$. Then $\angle OBA=$ 1

**(a)** $48^\circ$ **(b)** $58^\circ$ **(c)** $38^\circ$ **(d)** $68^\circ$

**46.** The height of a parallelogram of area $184$ cm$^2$ and base $23$ cm is (cm): 1

**(a)** 9 **(b)** 10 **(c)** 7 **(d)** 8

**47.** A die is rolled. The probability of getting a prime number or a multiple of 3 is: 1

**(a)** $\frac{1}{3}$ **(b)** $\frac{3}{4}$ **(c)** $\frac{2}{3}$ **(d)** $\frac{1}{2}$

**48.** The 4th and 9th terms of an AP are $14$ and $4$. Its 15th term is: 1

**(a)** $-7$ **(b)** $-9$ **(c)** $-6$ **(d)** $-8$

**49.** For the expression $n^2 + n + 107$, the smallest natural number $n$ for which the value is NOT prime is: 1

**(a)** $4$ **(b)** $5$ **(c)** $2$ **(d)** $3$

**50.** The average of 38 numbers is $40$. The average of all 57 numbers (including 19 more numbers) is $65$. The average of the 19 additional numbers is: 1

**(a)** $117$ **(b)** $115$ **(c)** $116$ **(d)** $114$

**END OF QUESTION PAPER** **Total: 50 Marks**

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